



Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 24-12-2025

Time: 09.00Am to 12.00Pm

GTM-04

Max. Marks: 300

KEY SHEET

MATHEMATICS

1	1	2	1	3	1	4	1	5	2
6	1	7	2	8	4	9	3	10	1
11	4	12	2	13	2	14	1	15	1
16	3	17	4	18	4	19	3	20	1
21	2890	22	9	23	119	24	29	25	12

PHYSICS

26	2	27	3	28	4	29	2	30	4
31	1	32	2	33	3	34	2	35	4
36	3	37	4	38	3	39	2	40	2
41	2	42	3	43	2	44	2	45	1
46	1150	47	90	48	5	49	30	50	3

CHEMISTRY

51	3	52	3	53	1	54	1	55	1
56	2	57	1	58	2	59	1	60	2
61	2	62	4	63	2	64	4	65	2
66	1	67	1	68	2	69	2	70	2
71	3	72	255	73	75	74	4	75	20



SOLUTION

MATHEMATICS

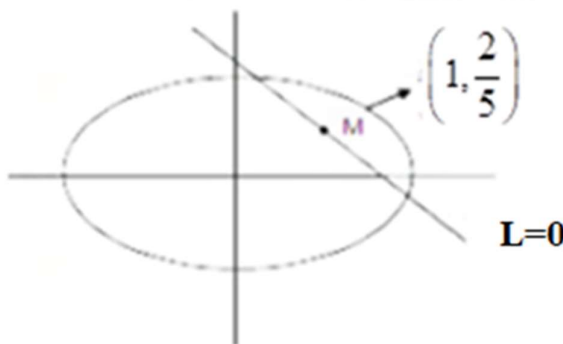
1. Given matrix is orthogonal

So, transpose of the given matrix will be its inverse

$$\begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is inverse of } f(x) \text{ which is equal to } f(-x)$$

$$\begin{aligned} f(x)f(y) &= \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix} = f(x+y) \end{aligned}$$

2. Equation of chord having midpoint $\left(1, \frac{2}{5}\right)$ is $S_1 = S_{11}$



$$\frac{x}{25} + \frac{y}{40} = \frac{1}{20}$$

$$L: \frac{x}{25} + \frac{y}{40} - \frac{1}{20} = 0$$

Let chord intersect ellipse at $A(x_1, y_1)$ and $B(x_2, y_2)$



$$AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = |x_1 - x_2| \sqrt{1 + \left(\frac{y_1 - y_2}{x_1 - x_2}\right)^2}$$

$$= |x_1 - x_2| \sqrt{1 + m_L^2} = |x_1 - x_2| \sqrt{1 + \frac{64}{25}}$$

$$AB = |x_1 - x_2| \frac{\sqrt{89}}{5}$$

Now, A and B are the point of intersection of ellipse and line.

$$\text{So, } \frac{x^2}{25} + \frac{(10-8x)^2}{25 \cdot 16} = 1$$

$$4x^2 - 8x - 15 = 0$$

$$x_1 + x_2 = 2 \text{ and } x_1 x_2 = \frac{-15}{4}$$

$$\Rightarrow |x_1 - x_2| = \sqrt{(x_1 + x_2)^2 - 4x_1 x_2} = \sqrt{19}$$

$$\text{So, } AB = |x_1 - x_2| \frac{\sqrt{89}}{5} = \frac{\sqrt{1691}}{5}$$

$$\int_0^1 \frac{\sqrt{3+x} - \sqrt{1+x}}{2} dx = \frac{1}{2} \left(\frac{2}{3} (3+x)^{\frac{3}{2}} - \frac{2}{3} (1+x)^{\frac{3}{2}} \right)_0^1$$

$$3. = \frac{1}{3} [9 - 2\sqrt{2} - 3\sqrt{3}]_0^1$$

$$= 3 - \frac{2}{3}\sqrt{2} - \sqrt{3} \quad a = 3, b = -\frac{2}{3}; c = -1$$

$$a = \lim_{x \rightarrow 0} \frac{\sqrt{1+\sqrt{1+x^4}} - \sqrt{2}}{x^4} = \lim_{x \rightarrow 0} \frac{1 + \sqrt{1+x^4} - 2}{x^4 (\sqrt{1+\sqrt{1+x^4}} + \sqrt{2})}$$

$$= \lim_{x \rightarrow 0} \frac{(1+x^4) - 1}{x^4 (\sqrt{1+x^4} + 1)} \cdot \frac{1}{(\sqrt{1+\sqrt{1+x^4}} + \sqrt{2})} = \frac{1}{4\sqrt{2}}$$

$$4. \quad b = \lim_{x \rightarrow 0} \frac{(1 - \cos^2 x)(\sqrt{2} + \sqrt{1 + \cos x})}{2 - (1 + \cos x)}$$

$$= \lim_{x \rightarrow 0} (1 + \cos x)(\sqrt{2} + \sqrt{1 + \cos x}) = (2)(2\sqrt{2}) = 4\sqrt{2}$$

$$\text{So } ab^3 = \frac{1}{4\sqrt{2}} \times 64(2\sqrt{2}) = 32$$



$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \frac{-a(x^2 - 7x + 12)}{b(x^2 - 7x + 12)} = \frac{-a}{b} \left[\begin{array}{l} \because x^2 - 7x + 12 > 0 \\ \text{If } x \rightarrow 3^- \end{array} \right]$$

$$5. \quad \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} 2^{\frac{\sin(x-3)}{x-3}} = 2 \quad \left[\begin{array}{l} \because [x] = 3 \\ \text{If } x \rightarrow 3^+ \end{array} \right]$$

$$f(3) = b$$

Now, $f(x)$ is continuous at $x=3$

$$\text{So, } \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = f(3) \Rightarrow \frac{-a}{b} = 2 = b$$

$$\text{So, } (a, b) = (-4, 2)$$

$$\therefore \left(\sum_{k=1}^{10} a_k \right)^2 = \sum_{k=1}^{10} a_k^2 + 2(\sum_{k \neq j} a_k a_j)$$

$$\Rightarrow (50)^2 = \sum_{k=1}^{10} a_k^2 + 2(1100)$$

$$6. \quad \Rightarrow \sum_{k=1}^{10} a_k^2 = 2500 - 2200 = 300$$

$$\therefore \sigma^2 = \left(\frac{\sum_{k=1}^{10} a_k^2}{10} \right) - \left(\frac{\sum_{k=1}^{10} a_k}{10} \right)^2$$

$$\Rightarrow \sigma^2 = \frac{300}{10} - \left(\frac{50}{10} \right)^2 = 30 - 25$$

$$\text{Standard deviation} = \sqrt{5}$$

$$7. \quad x^2 + y^2 - 4x - 16y + 64 = 0$$

$$\text{Centre } (2, 8) \text{ radius} = 2$$

shortest distance will lie along common normal

normal at $P(t)$ to parabola is

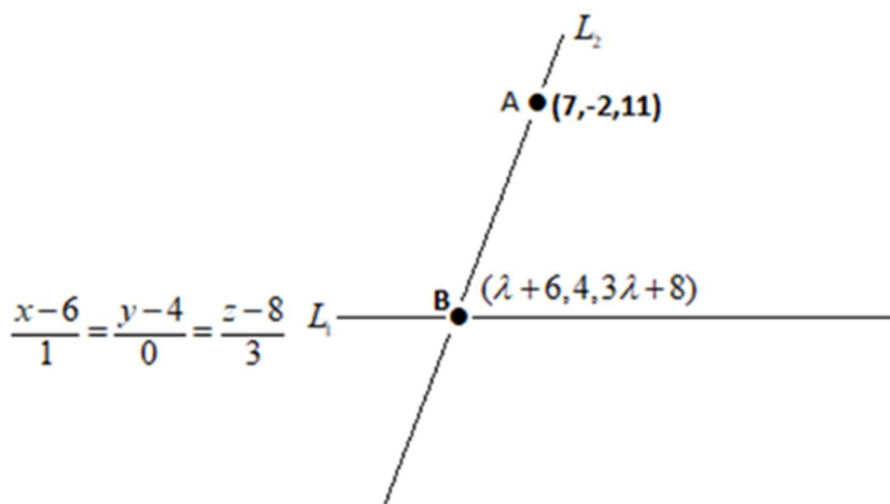
$$\therefore y + tx = 2t + t^3$$

Since it will pass through $(2, 8)$

$$\Rightarrow 8 + 2t = 2t + t^3 \quad t = 2$$

$$p(t) \equiv p(t^2, 2t) \equiv P(4, 4)$$

$$\therefore d = \sqrt{4 + 16} = \sqrt{20} \therefore d^2 = 20$$



8.

Equation of line passing through A is

$$L_2: \frac{x-7}{2} = \frac{y+2}{-3} = \frac{z-11}{6}$$

Let L_2 intersect line L_1 at point B

Let B is $(\lambda+6, 4, 3\lambda+8)$ $\left[\text{Take } \frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3} = \lambda \right]$

Since B satisfies line L_2

$$\text{So, } \frac{\lambda+6-7}{2} = \frac{6}{-3} = \frac{3\lambda+8-11}{6} \Rightarrow \frac{\lambda-1}{2} = -2 = \frac{3\lambda-3}{6}$$

$$\text{So, } \lambda = -3 \Rightarrow B \equiv (3, 4, -1)$$

$$AB = \sqrt{16+36+144} = 14$$

9.

$$L_1 \text{ is } \frac{x-4}{1} = \frac{y+1}{2} = \frac{z}{-3}$$

$$L_1: \vec{a}_1 = (4, -1, 0) \quad \vec{b}_1 = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$L_2 \text{ is } \frac{x-\lambda}{2} = \frac{y+1}{4} = \frac{z-2}{-5}$$



$$L_2 \xrightarrow[\vec{a}_2]{\vec{b}_2 = 2\hat{i} + 4\hat{j} - 5\hat{k}}$$

$(\lambda, -1, 2)$

$$\text{Shortest Distance} = \frac{|(\vec{a}_1 - \vec{a}_2) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$\text{where } \vec{a}_1 = 4\hat{i} - \hat{j}, \vec{a}_2 = \lambda\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{b}_1 = \hat{i} + 2\hat{j} - 3\hat{k}, \vec{b}_2 = 2\hat{i} + 4\hat{j} - 5\hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = 2\hat{i} - \hat{j}$$

$$\text{Shortest Distance} = \frac{|2\lambda - 8|}{\sqrt{5}} = \frac{6}{\sqrt{5}}$$

$$|2\lambda - 8| = 6$$

$$\lambda = 1, 7$$

$${}^{n-1}C_r = (k^2 - 8) \frac{n}{r+1} {}^{n-1}C_r \Rightarrow k^2 - 8 = \frac{r+1}{n}$$

$$\therefore n-1 \geq r \Rightarrow n \geq r+1$$

$$10. \text{ So, } 0 < \frac{r+1}{n} \leq 1$$

$$0 < k^2 - 8 \leq 18 < k^2 \leq 9$$

$$2\sqrt{2} < k \leq 3 \text{ or } -3 \leq k < -2\sqrt{2}$$

11.

$$\frac{dx}{dt} + ax = 0 \Rightarrow \frac{dx}{x} = -adt$$

$$\Rightarrow \log x = -at + c_1$$

$$\text{at } t = 0, x = 2$$

$$\therefore c_1 = \log 2$$

$$\Rightarrow \log x = -at + \log 2 \Rightarrow x = 2e^{-at}$$

$$\frac{dy}{dt} + by = 0 \Rightarrow \frac{dy}{y} = -bdt$$

$$\Rightarrow \log y = -bt + c_2$$

$$\text{at } t = 0, y = 1$$

$$\therefore c_2 = 0$$

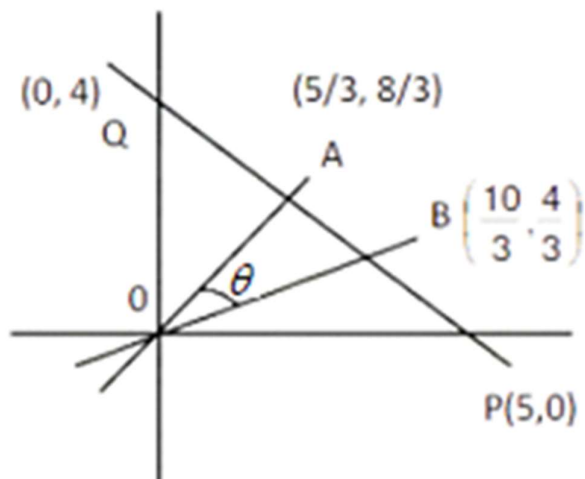
$$\Rightarrow \log y = -bt \Rightarrow y = e^{-bt}$$



$$\because 3y(1) = 2x(1) \Rightarrow 3e^{-b} = 2.2e^{-a} \Rightarrow e^{a-b} = \frac{4}{3}$$

$$\because x(t) = y(t) \Rightarrow 2e^{-at} = e^{-bt} \Rightarrow 2 = e^{(a-b)t} \Rightarrow 2 = \left(\frac{4}{3}\right)^t \Rightarrow t = \log_{\frac{4}{3}} 2$$

12. $\vec{a} \cdot ((\vec{c} \times \vec{b}) - \vec{b} - \vec{c}) = \vec{a} \cdot (\vec{c} \times \vec{b}) - \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c}$
 $\Rightarrow (\vec{a} \times \vec{c}) \cdot \vec{b} - 0 - 3 = \vec{b} \cdot \vec{b} - 3 = |\vec{b}|^2 - 3 = 27 - 3 = 24$



13.

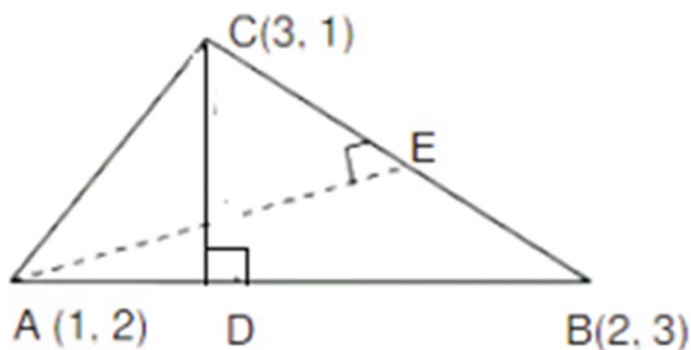
$$A\left(\frac{5}{3}, \frac{8}{3}\right), B\left(\frac{10}{3}, \frac{4}{3}\right)$$

$$m_{OA} = m_2 = \frac{8}{5}$$

$$m_{OP} = m_1 = \frac{2}{5}$$

θ is the angle between the lines L_1 and L_2 then

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \left| \frac{\frac{8}{5} - \frac{2}{5}}{1 + \frac{8}{5} \times \frac{2}{5}} \right| = \left| \frac{\frac{6}{5}}{1 + \frac{16}{25}} \right| = \frac{\frac{6}{5}}{\frac{25+16}{25}} = \frac{6 \times 5}{25+16} = \frac{30}{41}$$



14.

$$m_{AB} = 1 \quad m_{BC} = -2$$

$$\because CD \perp AB \therefore AE \perp BC$$

$$m_{CD} = -1 \quad m_{AE} = \frac{1}{2}$$

$$\text{Equation of CD is } y-1=-1(x-3) \quad x+y=4 \quad \dots\dots (1)$$

$$\text{Equation of AE is } y-2=\frac{1}{2}(x-1) \quad x-2y=-3 \quad \dots\dots (2)$$

Solving (1) and (2) we get co-ordinates of orthocenter is $H\left(\frac{5}{3}, \frac{7}{3}\right)$

$$\text{Now } I_1 = \int_{\frac{5}{3}}^{\frac{7}{3}} x \sin(4x - x^2) dx = \int_{\frac{5}{3}}^{\frac{7}{3}} x \sin(x(4-x)) dx \quad \dots\dots (3)$$

$$I_1 = \int_{\frac{5}{3}}^{\frac{7}{3}} (4-x) \sin((4-x)x) dx \quad \dots\dots (4) \quad (\text{king's property})$$

$$\text{Adding (3) and (4) we get } 2I_1 = \int_{\frac{5}{3}}^{\frac{7}{3}} 4 \sin(4x - x^2) dx$$

$$2I_1 = 4I_2$$

$$I_1 = 2I_2$$

$$\frac{I_1}{I_2} = 2$$

$$\frac{36I_1}{I_2} = 72$$

$$15. \quad \text{Let } E_1 \equiv (1-3x+10x^2)^n, E_2 \equiv (1+x^2)^n$$



Let $A =$ sum of all the coefficient of expansion $E_1 = E_1(1)$

$$\Rightarrow A=8$$

$B =$ sum of all the coefficient of expansion $E_2 = E_2(1) \Rightarrow B=2$

$$A = 8, B = 2, A=B^3$$

Alternate

Put $x=1$

$$(1-3+10)^n = A, (1+1)^n = B$$

$$8^n = A, 2^n = B$$

$$(2^n)^3 = A \Rightarrow A = B^3$$

16. Sequences : 4,9,14..... are in A.P. with C.D = 5

and 3,6,9,..... are in A.P. with C.D = 3

$$\text{Given } T_{25}=124 \text{ and } T_{37}=37 \times 3=111$$

Clearly first common term =9

Common difference = L.C.M. (5, 3) = 15

Numer of common terms of sequence of A.P. is

$$= 9 + (n - 1)15$$

$$= 15n - 6$$

$$\therefore 15n-6 \leq 111 \quad 15n \leq 117 \quad n \leq 7.8$$

Number of common terms = 7

17. Reflexive : Let $A \in M$ (Subsets of 'S')

$\phi \cap \phi = \phi$ is not satisfy given relation

So R is not reflexive

Symmetric : If $(A,B) \in R, A \cap B \neq \phi \Rightarrow B \cap A \neq \phi$

$\Rightarrow (B,A) \in R$ So R is symmetric

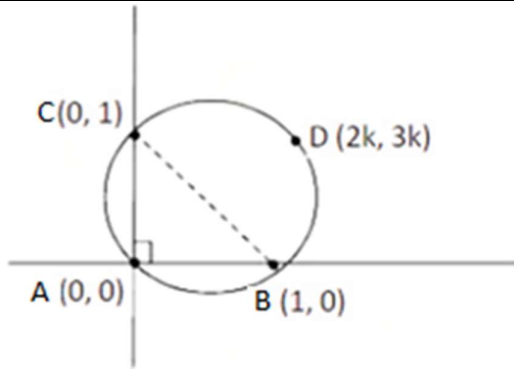
Transitive : If (A,B) and $(B,C) \in R \Rightarrow A \cap B \neq \phi$ and $B \cap C \neq \phi$

$\Rightarrow A \cap C$ is not necessarily ϕ set. Hence $A \cap C$ not necessarily belongs to R

So R is not a transitive relation

Eg. $A = \{1, 2\}, B = \{3, 4\}, C = \{2, 5\}$

$A \cap B = \phi, B \cap C = \phi$ but $A \cap C = \{2\} \neq \phi$



18.

Equation of a circle when BC is a diameter is $(x-1)(x-0)+(y-0)(y-1)=0$

In satisfy $D(2k, 3k)$

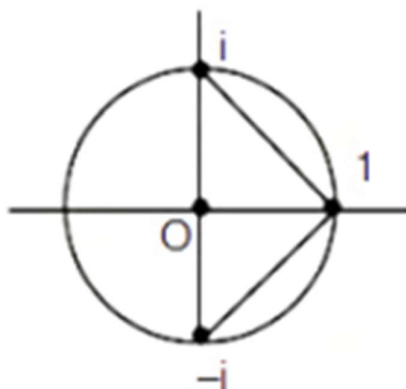
$$(2k-1)(2k)+(3k)(3k-1)=0$$

$$4k^2 - 2k + 9k^2 - 3k = 0$$

$$13k^2 - 5k = 0 \Rightarrow k = 0, \frac{5}{13}$$

When $k = 0$ point $D(0, 0)$

So value of $k = \frac{5}{13}$

19. $n(S)=1$ 

20. Largest prime factors of two different natural numbers can be same so it is many – one. Example $f(6)=3, f(12)=3$

As range of $f(x)$ is set of prime numbers so range of $f(x)$ is proper subset of Co – domain so f is into.

Alternate :

Given, $f: \mathbb{N} - \{1\} \rightarrow \mathbb{N}$



highest prime factor of n

Clearly, f is not onto because a set of natural numbers is not equal to the set of prime numbers.

The range of ' f ' is set prime numbers.

$$21. \quad f(x) - f(y) \geq \log_e \left(\frac{x}{y} \right) + x - y \dots (1)$$

Interchange x and y

$$f(y) - f(x) \geq \log_e \left(\frac{y}{x} \right) + y - x \Rightarrow f(x) - f(y) \leq \log_e \left(\frac{x}{y} \right) + x - y \dots (2)$$

$$\text{From (1) and (2)} \Rightarrow f(x) - f(y) = \log_e \left(\frac{x}{y} \right) + x - y$$

$$\lim_{x \rightarrow y} \left(\frac{f(x) - f(y)}{x - y} \right) = \lim_{x \rightarrow y} \left(\frac{\ln(x) - \ln(y)}{x - y} + 1 \right)$$

$$f'(y) = \frac{1}{y} + 1 \quad f' \left(\frac{1}{n^2} \right) = n^2 + 1$$

$$\sum_{n=1}^{20} f' \left(\frac{1}{n^2} \right) = \sum_{n=1}^{20} (n^2 + 1) = \frac{(20)(21)(41)}{6} + 20 = 2870 + 20 = 2890$$

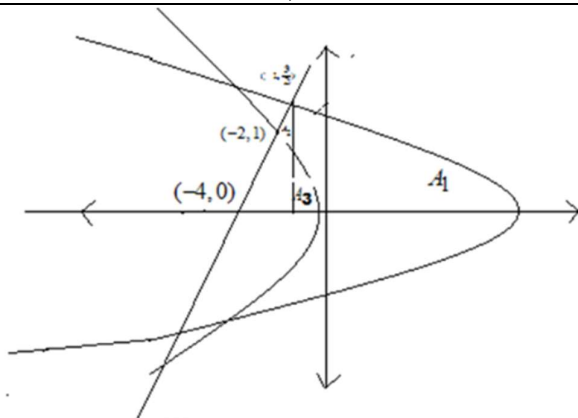
$$8 = 3 + \frac{1}{4}(3+p) + \frac{1}{4^2}(3+2p) + \frac{1}{4^3}(3+3p) + \dots \dots (1)$$

$$\frac{(1)}{4} \Rightarrow 2 = \frac{3}{4} + \frac{3+p}{4^2} + \frac{3+2p}{4^3} \dots (2)$$

$$22. \quad (1) - (2) \quad 6 = 3 + \frac{p}{4} + \frac{p}{4^2} + \dots$$

$$3 = \frac{p}{4} + \frac{p}{4^2} + \dots$$

$$3 = \frac{\frac{p}{4}}{1 - \frac{1}{4}} \Rightarrow p = 9$$



23.

$$A = \int_0^1 [(8-4y^2) - (-2y^2)] dy + \int_1^3 [(8-4y^2) - (2y-4)] dy$$

$$= \left[8y - \frac{2y^3}{3} \right]_0^1 + \left[12y - y^2 - \frac{4y^3}{3} \right]_1^3 = \frac{107}{12} = \frac{m}{n}$$

$$\therefore m+n=119$$

$$\frac{dy}{dx} = -\left(\frac{2x+3y-2}{4x+6y-7} \right) \dots (1)$$

$$\text{let } 2x+3y=t \Rightarrow 2+3 \cdot \frac{dy}{dx} = \frac{dt}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{3} \left(\frac{dt}{dx} - 2 \right)$$

$$\text{From (1)} \quad \frac{1}{3} \left(\frac{dt}{dx} - 2 \right) = -\left(\frac{t-2}{2t-7} \right) \Rightarrow \frac{dt}{dx} = \frac{-3t+6}{2t-7} + 2$$

$$\Rightarrow \frac{dt}{dx} = \frac{t-8}{2t-7} \Rightarrow \frac{2t-7}{t-8} dt = dx \Rightarrow \int \frac{2(t-8)+9}{t-8} dt = \int dx$$

$$\Rightarrow \int \left(2 + \frac{9}{t-8} \right) dt = x + c \Rightarrow 2t + 9 \ln(t-8) = x + c$$

$$\Rightarrow 2(2x+3y) + 9 \ln(2x+3y-8) = x + c$$

$$\text{Since } y(0)=3 \Rightarrow c=18$$

$$\Rightarrow (x+2y) + 3 \ln(2x+3y-8) = 6$$

$$\alpha + 2\beta + 3\gamma = 29$$

$$a = p(X=3) = \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} = \frac{25}{216}$$

$$b = p(X \geq 3) = \frac{5^2}{6^3} + \frac{5^3}{6^4} + \frac{5^4}{6^5} + \dots = \frac{25}{36}$$

$$25. \quad c = \frac{p(X \geq 6)}{p(X > 3)} = \frac{\frac{5^5}{6^6} + \frac{5^6}{6^7} + \frac{5^7}{6^8} + \dots}{\frac{5^3}{6^4} + \frac{5^4}{6^5} + \frac{5^5}{6^6} + \dots} = \frac{25}{36}$$

$$\frac{b+c}{a} = \frac{50/36}{25/216} = 12$$

**PHYSICS**

$$a = -\omega^2 x$$

$$\Rightarrow a \propto x$$

$$\Rightarrow F_R \propto x$$

$$V = \omega \sqrt{A^2 - x^2}$$

$$\Rightarrow V_{\max} = A\omega \text{ (at mean position)}$$

$$\Rightarrow |a_{\max}| = \omega^2 A \text{ (at extreme position)}$$

$$N = \frac{N_0}{2^m}$$

$$m = \frac{t}{t_{1/2}} = \frac{15}{5} = 3$$

$$N = \frac{N_0}{(2)^3}$$

$$\frac{N}{N_0} = \frac{1}{8}$$

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mE}}$$

$$\lambda^1 = \frac{h}{\sqrt{\frac{2mE}{4}}} = 2\lambda$$

$$\lambda = a \sin \theta$$

$$\Rightarrow a = \frac{600 \times 10^{-9}}{\sin 30^\circ} = 6 \times 10^{-7} \times 2$$

$$\Rightarrow a = 12 \times 10^{-7} = 1.2 \mu\text{m}$$

$$\lambda_{\text{mw}} > \lambda_{\text{IR}} > \lambda_{\text{UV}} > \lambda_{\text{x-ray}}$$

(A)-(IV), (B)-(I), (C)-(II), (D)-(III)

$$R_{\text{eq}} = \frac{20 \times 10}{20 + 10} = \frac{20}{3} \Omega$$

$$\Rightarrow I = \frac{10 \times 3}{20} = \frac{3}{2} = 1.5 \text{ A}$$

Area under P-V curve gives the amount of work.

$$W = \frac{1}{2} \times 2 \times 300 = 300 \text{ J}$$

$$d_{\max} = \sqrt{2RH_T} + \sqrt{2RH_R}$$

$$= \sqrt{\frac{2 \times 6400 \times 180}{1000}} + \sqrt{\frac{2 \times 6400 \times 245}{1000}}$$

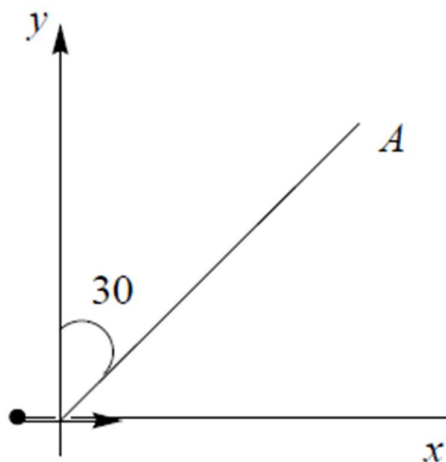
$$= 8 \times 6 + 8 \times 7 = 104 \text{ km}$$

$$\vec{r} = 10t\hat{i} + 15t^2\hat{j} + 7\hat{k}$$

$$\vec{V} = \frac{d\vec{r}}{dt} = 10\hat{i} + 30t\hat{j} + 0$$

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} = 30\hat{j}$$

Hence, force is along positive y-axis.



35.

$$A_y = A \cos 30^\circ = 2\sqrt{3}$$

$$\Rightarrow A = \frac{2\sqrt{3}}{\cos 30^\circ}$$

x - component :

$$A_x = A \sin 30^\circ = \left(\frac{2\sqrt{3}}{\cos 30^\circ} \right) \sin(30^\circ)$$

$$A_x = 2\sqrt{3} \tan 30^\circ = 2$$

36. For voltmeter:

$$R = \frac{V}{i_g} - G = 50000 - 54 = 49946 \approx 50 \text{ k}\Omega$$

For Ammeter:

$$r = \frac{i_g G}{i - i_g} = \frac{54}{9} = 6 \Omega$$

37. In series combination, the equivalent resistance is always greater than the least resistance in the combination.

Resistivity of a material is temperature dependent.

$$\rho_t = \rho_0(1 + \alpha t)$$

$$38. E_{\text{equatorial}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3}$$

$$\Rightarrow E \propto \frac{1}{r^3}$$

39. For a given material, $Y = \text{constant}$

$$\Rightarrow \frac{fL}{\pi r^2 l} = \frac{2f \times 2L}{\pi (4r^2) e_2}$$

$$\Rightarrow e_2 = 1$$

40. From conservation of energy,



$$\frac{-GMm}{2R} + 0 = \frac{-GMm}{R} + \frac{1}{2}mv^2$$

$$\Rightarrow v^2 = \frac{GM}{R} = gR$$

$$\Rightarrow v = \sqrt{gR}$$

$$V = \lambda^a g^b \rho^c$$

$$\Rightarrow LT^{-1} = (L)^a (LT^{-2})^b (ML^{-3})^c$$

$$\Rightarrow LT^{-1} = L^{a+b-3c} T^{-2b} M^c$$

$$41. \therefore c = 0, 2b = 1 \Rightarrow b = \frac{1}{2},$$

$$a + b - 3c = 1$$

$$\Rightarrow a = \frac{1}{2}$$

$$V = \frac{dx}{dt} = \frac{d}{dt}(5t^2 - 4t + 5)$$

$$42. V = 5(2t) - 4(1) + 0$$

$$V = 10t - 4$$

$$V_{t=2s} = 10 \times 2 - 4 = 16 \text{ m/s}$$

$$43. K.E = \frac{3}{2}kT \text{ is same for all gases}$$

$$\frac{KE_{He}}{KE_{Ar}} = \frac{\frac{3}{2}kT}{\frac{3}{2}kT} = 1$$

$$R_{eq} = \frac{6 \times 6}{12} = 3 \Omega$$

$$44. i = i_1 + i_2 \Rightarrow i_1 = i_2 = \frac{3}{2} \text{ A}$$

$$V_A = \frac{3}{2} \times 4 = 6 \text{ V}; V_B = \frac{3}{2} \times 2 = 3 \text{ V}$$

$$V_{AB} = V_A - V_B = 6 - 3 = 3 \text{ V}$$

$$f_c = F_g$$

$$45. \Rightarrow m\omega^2 = \frac{Gm^2}{4a^2}$$

$$\Rightarrow \omega = \sqrt{\frac{Gm}{4a^3}}$$

$$46. W = \int F dx = \int_2^4 5x dx = 5 \left(\frac{x^2}{2} \right)_2^4$$

$$\Rightarrow W = \frac{5}{2}(16 - 4) = 30 \text{ J}$$



$$\Delta P = h \rho g + \frac{2T}{r}$$

$$47. \quad = 0.1 \times 10^3 \times 10 + \frac{2 \times 0.075}{1 \times 10^{-3}}$$

$$= 1000 + 150 = 1150 \text{ Pa}$$

$$f_0 = \frac{v}{2L}; \mu = \frac{m}{L}$$

$$48. \quad f_0 = \frac{\mu v}{2m} \Rightarrow v = \frac{2mf_0}{\mu}$$

$$\Rightarrow v = \frac{2 \times 18 \times 50}{20} = 90 \text{ m/s}$$

$$n_A = 2; n_B = 3; n_C = 4$$

$$\frac{1}{\lambda_1} = RZ^2 \left(\frac{1}{9} - \frac{1}{16} \right) = RZ^2 \times \frac{7}{16 \times 9}$$

$$49. \quad \frac{1}{\lambda_2} = RZ^2 \left(\frac{1}{4} - \frac{1}{9} \right) = RZ^2 \times \frac{5}{9 \times 4}$$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{7}{5 \times 4} = \frac{7}{4n}$$

$$\Rightarrow n = 5$$

$$\mu = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \sqrt{2} \sin 30^\circ = \sin\left(\frac{\delta_{\min} + 60^\circ}{2}\right)$$

$$50. \quad \Rightarrow \frac{1}{\sqrt{2}} = \sin\left(\frac{\delta_{\min} + 60^\circ}{2}\right)$$

$$\Rightarrow 45^\circ = \left(\frac{\delta_{\min} + 60^\circ}{2}\right)$$

$$\Rightarrow \delta_{\min} = 90^\circ - 60^\circ = 30^\circ$$



CHEMISTRY

$$\begin{array}{ccc} \text{C} & : \text{H} & \text{O} \\ \frac{54.2}{12} & : 9.2 & : \frac{36.6}{16} \end{array}$$

51. $4.516 : 9.2 : 2.287$

$$\frac{4.516}{2.287} : \frac{9.2}{2.287} : \frac{2.287}{2.287}$$

$$1.97 : 4.02 : 1$$

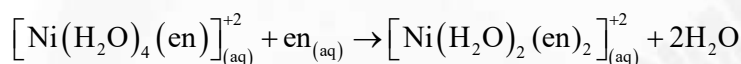
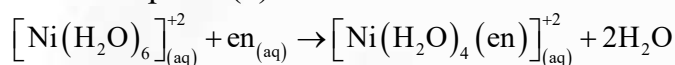
$\text{C}_2\text{H}_4\text{O} \Rightarrow$ Empirical formula

E.F. mass = 24 + 4 + 16 = 44 and molar mass = 132

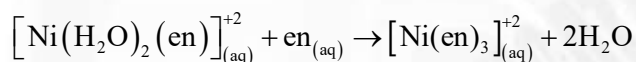
Hence molecular formula = $(\text{C}_2\text{H}_4\text{O})_3$



Correct Option (3)



52. Pale Blue



Blue / purple

53. For $3d^4$

If ligand is SFL : SFL : $t_{2g}^4 e_g^0$ (Low spin)

If ligand is WFL: WFL: $t_{2g}^3 e_g^1$ (High spin)

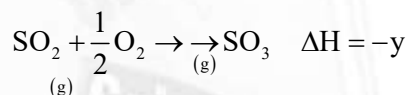
54. (B) $-\text{CN}$ is meta directing But $-\text{OH}$ is ortho / para directing.

(E) Halides are deactivating groups.

55. Order of I.E. (in KJ/mol) :



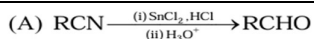
$$1086 \quad 786 \quad 761 \quad 708 \quad 715$$



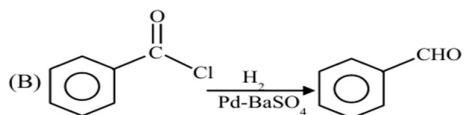
56. $\Delta H_r = (\Delta H_f)_{\text{SO}_3} - (\Delta H_f)_{\text{SO}_2}$

$$-y = -2x - (\Delta H_f)_{\text{SO}_2}$$

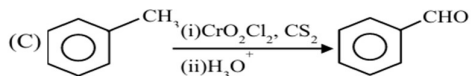
$$(\Delta H_f)_{\text{SO}_2} = y - 2x$$



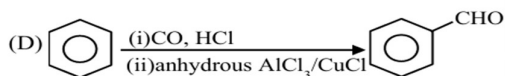
Stephen reaction



Rosenmund reduction



Etard reaction



57.

Gatterman-Koch reaction



58.

59.

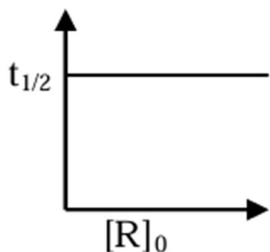
	List-I		List-II
(A)	Adenine	(III)	
(B)	Cytosine	(IV)	
(C)	Thymine	(II)	
(D)	Uracil	(I)	

60. The IE_4 is suddenly very high therefore element must have 3 valence e^- (s) and it belong to group 13.



61. For strongest reducing agent
Reduction potential should be lowest
Hence Cr is the strongest reducing agent.
Options (2)

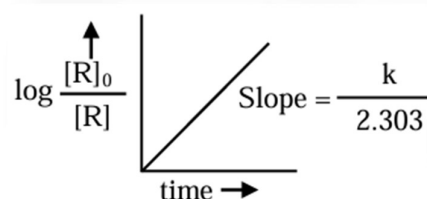
62. For first order reaction $t_{1/2} = \frac{\ln 2}{k}$



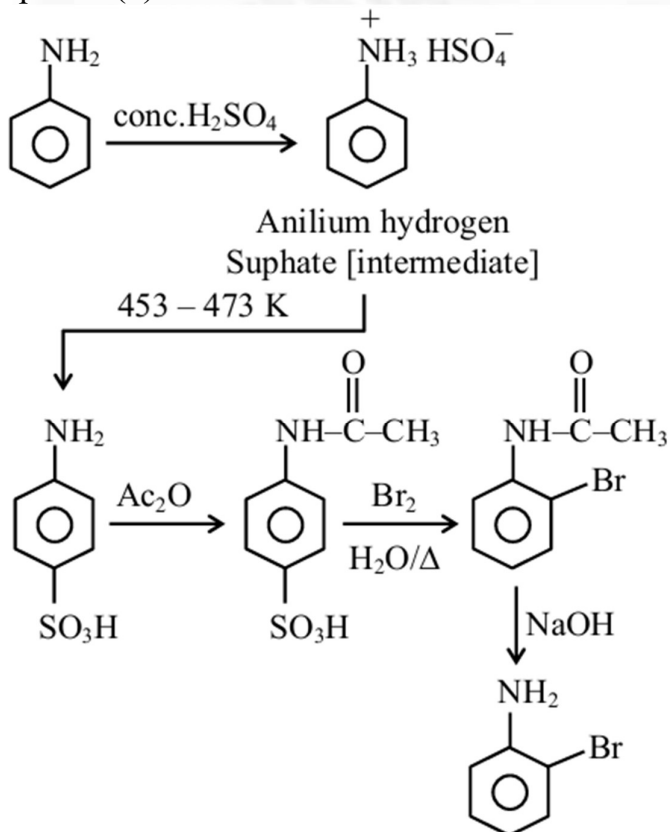
For first order reaction

$$\log \frac{[R]_0}{[R]} = \frac{1}{2.303} kt$$

$$\log \frac{[R]_0}{[R]} = \left(\frac{k}{2.303} \right) \times t$$



Options (4)

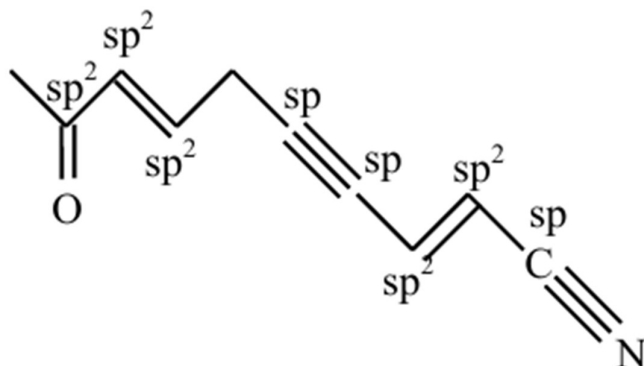


63.



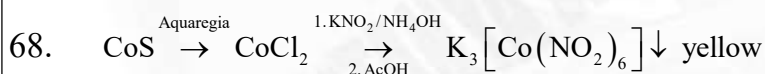
64. In a hydrogen atom, the energy of orbitals in a shell depends only on 'n'

$$\therefore E_{3p_x} = E_{3d_{x^2-y^2}} = E_{3d_{z^2}}$$



65. Number of sp and sp^2 hybridised carbon atom are 3 and 5.
66. Higher the number of milli moles of acid or base reacted higher will be temperature rise.
 Option (4) n_{acid} or n_{base} reacted = 30 m mol
 Option (2) n_{acid} or n_{base} reacted = 30 m mol
 But less energy will be released by neutralisation reaction of weak acid hence option (2) can not be correct.
 Option (3) $\Rightarrow$ 20 m mol
 Option (4) $\Rightarrow$ 25 m mol
 Hence Correct Option (1)

67. Due to resonance bond length is identical in ozone.
 Therefore statement I is true and statement II is false.



69. $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$

Concentration of H_2 and I_2 decreases untill equilibrium condition and concentration of HI increases till equilibrium condition and after equilibrium concentration of all the reactant and products remain constant.

Correct option (2)

$$\text{Sc}^{+3} = 3d^0 \quad \therefore \mu_{\text{spin}} = 0$$

$$\text{V}^{+2} = 3d^3 \quad \therefore \mu_{\text{spin}} = 3.87 \text{ B.M.}$$

$$\text{Ni}^{+2} = 3d^8 \quad \therefore \mu_{\text{spin}} = 2.84 \text{ B.M.}$$

$$\text{Ti}^{+3} = 3d^1 \quad \therefore \mu_{\text{spin}} = 1.73 \text{ B.M.}$$

71. Mass of carbon = $\frac{80 \times 90}{100} = 72 \text{ gm}$

$$\text{Number of C-atoms} = \frac{72}{12} = 6$$

$$\text{Mass of hydrogen} = \frac{80 \times 10}{800} = 8 \text{ gm}$$

$$\text{Number of H-atoms} = \frac{8}{1} = 8$$

So molecular formula C_6H_8

$$\text{D.U.} = 6 + 1 - 8/2 = 7 - 4 = 3$$



72.
$$\% \text{ Bromine} = \frac{\text{Molar Mass of Bromine}}{\text{Molar Mass of Silver bromide}} \times \frac{\text{Weight of AgBr}}{\text{Weight of sample}} \times 100$$

$$= \frac{80}{188} \times \frac{0.165}{0.25} \times 100 = \frac{4800}{188} = 25.53 = 255 \times 10^{-1}$$



$$i = \frac{\text{normal molar mass}}{\text{observed molar mass}}$$

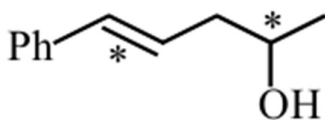
$$i = \frac{164}{65.6}$$

73.
$$1 + (3-1)\alpha = \frac{164}{65.6}$$

$$2\alpha = \frac{98.4}{65.6}$$

$$\alpha = 0.75$$

percent dissociation = 75%



74. n (stereogenic unit) = 2, $2^2 = 4$ stereoisomers are possible.

$$K = \sqrt{\frac{K_1 K_3}{K_2}}$$

75.
$$A.e^{-E_a/RT} = \sqrt{\frac{A_1 e^{-E_{a1}/RT} \times A_3 e^{-E_{a3}/RT}}{A_2 e^{-E_{a2}/RT}}}$$

By comparing exponential term

$$\frac{E_a}{RT} = \frac{1}{2} \times \left(\frac{E_{a1}}{RT} + \frac{E_{a3}}{RT} - \frac{E_{a2}}{RT} \right)$$

$$E_a = (E_{a1} + E_{a3} - E_{a2}) / 2$$

$$E_a = (60 + 10 - 30) / 2 = 20 \text{ kJ mol}^{-1}$$

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